

ECO 329 ECONOMIC STATISTICS (36110)

Lecture Notes

Fall 2025

7.3 Two-Sample Hypothesis Testing

In the section, unless specified otherwise, let $\{X_i\}_{i=1}^{n_X}$ and $\{Y_i\}_{i=1}^{n_Y}$ be two random samples that are independent with one another and with population means μ_X, μ_Y as well as population variances σ_X^2, σ_Y^2 respectively (and drawn from the populations specified (if ever) in each of the following particular situations).

Proposition 7.3.1.

$$E(\bar{X} - \bar{Y}) = \mu_X - \mu_Y \text{ and } V(\bar{X} - \bar{Y}) = \frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}.$$

- Proof idea: Direct calculation by property of linear combinations.

Method 7.3.2. Test about mean difference of two normal populations $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$ with σ_X^2 and σ_Y^2 both known.

1. Here, following our routine applied in the previous one-sample tests, we work on the following three pairs of hypotheses:
 - (a) $H_0 : \mu_X - \mu_Y = \Delta\mu, H_A : \mu_X - \mu_Y \neq \Delta\mu,$
 - (b) $H_0 : \mu_X - \mu_Y \leq \Delta\mu, H_A : \mu_X - \mu_Y > \Delta\mu,$ and
 - (c) $H_0 : \mu_X - \mu_Y \geq \Delta\mu, H_A : \mu_X - \mu_Y < \Delta\mu.$

2. By property of random sample and linear combination of normal r.v.'s,

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} \sim N(0, 1).$$

- With the test statistic and the sampling distribution above, we can
 - proceed with the test by following the corresponding subsequent procedures listed in Method 7.2.1, and
 - construct CIs for the mean difference accordingly.

Method 7.3.3. Test about mean difference of two populations with means μ_X, μ_Y , unknown variances σ_X^2, σ_Y^2 , and large sample sizes.

1. Similar to the previous situation, we work on the following three pairs of hypotheses:

- (a) $H_0 : \mu_X - \mu_Y = \Delta\mu$, $H_A : \mu_X - \mu_Y \neq \Delta\mu$,
- (b) $H_0 : \mu_X - \mu_Y \leq \Delta\mu$, $H_A : \mu_X - \mu_Y > \Delta\mu$, and
- (c) $H_0 : \mu_X - \mu_Y \geq \Delta\mu$, $H_A : \mu_X - \mu_Y < \Delta\mu$.

2. By a CLT, approximately (or, rigorously, in the limit),

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n_X} + \frac{S_Y^2}{n_Y}}} \sim N(0, 1).$$

- With the test statistic and the sampling distribution above, we proceed with the test by following the corresponding subsequent procedures listed in Method 7.2.2.
- Again, we can directly replace S^2 's with σ^2 's if the population variances are known.
- However, in cases for Bernoulli populations, we can no longer follow the large-sample method analogous to the one-sample situation because the hypotheses here are only about the difference between the success probabilities of the two populations, while the variance associated with the test statistic yielded from a CLT is

$$\frac{\mu_X(1 - \mu_X)}{n_X} + \frac{\mu_Y(1 - \mu_Y)}{n_Y},$$

which depends on the exact (separate) values of the success probabilities.

- Therefore, the test statistic and sampling distribution in a two-sample Bernoulli case would have to be

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\bar{X}(1-\bar{X})}{n_X} + \frac{\bar{Y}(1-\bar{Y})}{n_Y}}} \sim N(0, 1).$$

Method 7.3.4. Test about mean difference of two normal populations $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$ with σ_X^2 and σ_Y^2 both unknown but assumed to be equal and small sample sizes.

- Because it is assumed that $\sigma_X = \sigma_Y = \sigma$, an estimator of the variance of the newly combined sample from X 's and Y 's is

$$S^2 = \frac{(n_X - 1)S_X^2 + (n_Y - 1)S_Y^2}{n_X + n_Y - 2},$$

and following the manner of Proposition 5.6,

$$\frac{(n_X + n_Y - 2)S^2}{\sigma^2} \sim \chi_{n_X + n_Y - 2}^2.$$

- Then, combined with

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sigma \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}} = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} \sim N(0, 1),$$

again following the manner of Proposition 5.6,

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}} \sim t_{n_X + n_Y - 2},$$

which are the test statistic and sampling distribution according to which we could do the test.

- (Not required) In cases where $\sigma_X \neq \sigma_Y$, a method, namely Welch-Satterthwaite method, gives the following test statistic and sampling distribution:

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n_X} + \frac{S_Y^2}{n_Y}}} \sim t_v,$$

where

$$v = \frac{(\frac{S_X^2}{n_X} + \frac{S_Y^2}{n_Y})^2}{\frac{(\frac{S_X^2}{n_X})^2}{n_X - 1} + \frac{(\frac{S_Y^2}{n_Y})^2}{n_Y - 1}}.$$

Definition 7.3.5. Let $K_1 \sim \chi_{k_1}^2$ and $K_2 \sim \chi_{k_2}^2$ be independent, then a r.v. F is said to follow a **F distribution** with k_1 and k_2 degrees of freedom, denoted as $F \sim F_{k_1, k_2}$, if

$$F = \frac{\frac{K_1}{k_1}}{\frac{K_2}{k_2}}.$$

- The pdf of F_{k_1, k_2} extends from 0 to $+\infty$ and is positively skewed, and its exact shape depends on the values of k_1 and k_2 .
- The $100(1 - \alpha)^{th}$ percentile of F_{k_1, k_2} is conventionally denoted as F_{α, k_1, k_2} - and such F_{α, k_1, k_2} 's are usually referred to as F critical values.

$$- F_{1-\alpha, k_2, k_1} = \frac{1}{F_{\alpha, k_1, k_2}}$$

Proposition 7.3.6. For two independent random samples $\{X_i\}_{i=1}^{n_X}$ and $\{Y_i\}_{i=1}^{n_Y}$ drawn from $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$ respectively,

$$\frac{\frac{S_X^2}{\sigma_X^2}}{\frac{S_Y^2}{\sigma_Y^2}} \sim F_{n_X - 1, n_Y - 1}.$$

- Proof: not required.

Method 7.3.7. Test about variance ratio of two normal populations $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$.

1. In this situation, we work on the following three pairs of hypotheses:

- (a) $H_0 : \frac{\sigma_X^2}{\sigma_Y^2} = \rho, H_A : \frac{\sigma_X^2}{\sigma_Y^2} \neq \rho,$
- (b) $H_0 : \frac{\sigma_X^2}{\sigma_Y^2} \leq \rho, H_A : \frac{\sigma_X^2}{\sigma_Y^2} > \rho,$ and
- (c) $H_0 : \frac{\sigma_X^2}{\sigma_Y^2} \geq \rho, H_A : \frac{\sigma_X^2}{\sigma_Y^2} < \rho.$

2. By Proposition 7.3.6, the test statistic and sampling distribution usable in this case is

$$\frac{\frac{S_X^2}{\sigma_X^2}}{\frac{S_Y^2}{\sigma_Y^2}} \sim F_{n_X-1, n_Y-1}.$$

3. For each pair of hypotheses, the rejection region is constructed in the following manner (with cv 's standing for critical values):

- (a) $H_0 : \frac{\sigma_X^2}{\sigma_Y^2} = \rho, H_A : \frac{\sigma_X^2}{\sigma_Y^2} \neq \rho.$

$$\begin{aligned} \frac{\alpha}{2} &= P\left(\frac{S_X^2}{S_Y^2} < cv_1 \mid \frac{\sigma_X^2}{\sigma_Y^2} = \rho\right) = P\left(\frac{S_X^2}{S_Y^2} > cv_2 \mid \frac{\sigma_X^2}{\sigma_Y^2} = \rho\right) \\ &= P\left(\frac{\frac{S_X^2}{\sigma_X^2}}{\frac{S_Y^2}{\sigma_Y^2}} < \frac{cv_1 \sigma_Y^2}{\sigma_X^2} \mid \frac{\sigma_X^2}{\sigma_Y^2} = \rho\right) = P\left(\frac{\frac{S_X^2}{\sigma_X^2}}{\frac{S_Y^2}{\sigma_Y^2}} > \frac{cv_2 \sigma_Y^2}{\sigma_X^2} \mid \frac{\sigma_X^2}{\sigma_Y^2} = \rho\right), \text{ constituting the} \end{aligned}$$

$$\text{rejection region: } (0, \rho F_{1-\frac{\alpha}{2}, n_X-1, n_Y-1}) \cup (\rho F_{\frac{\alpha}{2}, n_X-1, n_Y-1}, +\infty).$$

- (b) $H_0 : \frac{\sigma_X^2}{\sigma_Y^2} \leq \rho, H_A : \frac{\sigma_X^2}{\sigma_Y^2} > \rho.$

$$\alpha = \max_{\rho' \leq \rho} P\left(\frac{S_X^2}{S_Y^2} > cv \mid \frac{\sigma_X^2}{\sigma_Y^2} = \rho'\right) = \max_{\frac{\sigma_X^2}{\sigma_Y^2} \leq \rho} P\left(\frac{\frac{S_X^2}{\sigma_X^2}}{\frac{S_Y^2}{\sigma_Y^2}} > \frac{cv \sigma_Y^2}{\sigma_X^2}\right), \text{ constituting the}$$

$$\text{rejection region: } \bigcap_{\rho' \leq \rho} (\rho' F_{\alpha, n_X-1, n_Y-1}, +\infty) = (\rho F_{\alpha, n_X-1, n_Y-1}, +\infty).$$

- (c) $H_0 : \frac{\sigma_X^2}{\sigma_Y^2} \geq \rho, H_A : \frac{\sigma_X^2}{\sigma_Y^2} < \rho.$

$$\alpha = \max_{\rho' \geq \rho} P\left(\frac{S_X^2}{S_Y^2} < cv \mid \frac{\sigma_X^2}{\sigma_Y^2} = \rho'\right) = \max_{\frac{\sigma_X^2}{\sigma_Y^2} \geq \rho} P\left(\frac{\frac{S_X^2}{\sigma_X^2}}{\frac{S_Y^2}{\sigma_Y^2}} < \frac{cv \sigma_Y^2}{\sigma_X^2}\right), \text{ constituting the}$$

$$\text{rejection region: } \bigcap_{\rho' \geq \rho} (0, \rho' F_{1-\alpha, n_X-1, n_Y-1}) = (0, \rho F_{1-\alpha, n_X-1, n_Y-1}).$$

- Meanwhile, we can also get the relevant $1 - \alpha$ confidence intervals (of $\frac{\sigma_X^2}{\sigma_Y^2}$) in each of the situations above.

(a) $\left[\frac{S_X^2}{S_Y^2 F_{\frac{\alpha}{2}, n_X-1, n_Y-1}}, \frac{S_X^2}{S_Y^2 F_{1-\frac{\alpha}{2}, n_X-1, n_Y-1}} \right].$

(b) $\left[\frac{S_X^2}{S_Y^2 F_{\alpha, n_X-1, n_Y-1}}, +\infty \right).$

(c) $\left(0, \frac{S_X^2}{S_Y^2 F_{1-\alpha, n_X-1, n_Y-1}} \right].$

Example 7.A: Normal Population, Known Variances

Consider a sports equipment manufacturer whose 2-lb dumbbells are with weights normally distributed with mean μ_X and variance 0.0016 (unit: lb). Also, its 4-lb dumbbells are with weights normally distributed with mean μ_Y and variance 0.0025 (unit: lb). A random sample of 2-lb dumbbells of sample size 32 and another random sample of 4-lb dumbbells of sample size 50 are obtained from the manufacturer with sample means $\bar{X} = 2.002$ and $\bar{Y} = 4.003$ (unit: lb), and the two samples are independent from one another.

Write down the null and the alternative hypotheses for a researcher who hopes to test that the weight of the 4-lb dumbbells is exactly heavier than that of the 2-lb dumbbells by 2 lbs. Can we reject the null at significance level $\alpha = 10\%$?

Note: We will need some of the following z critical values in this problem: $z_{0.1} = 1.2816$, $z_{0.05} = 1.6449$, $z_{0.025} = 1.96$.

Example 7.B: Large Sample, Unknown Population

Consider two assembly line of a chip manufacturer. One of the assembly lines is old, whose hourly output is with mean μ_X and variance σ_X^2 . The other assembly is new, whose hourly output is with mean μ_Y and variance σ_Y^2 . The population means and variances above are all unknown. A random sample of hourly outputs of the old assembly line with large sample size 300 is obtained from the assembly line, which is with sample mean $\bar{X} = 1800$ and sample variance $S_X^2 = 27$, and another random sample of hourly outputs of the new assembly line with large sample size 400 is obtained from the assembly line, which is with sample mean $\bar{Y} = 2000$ and sample variance $S_Y^2 = 64$.

Write down the null and the alternative hypotheses for a researcher who hopes to test that the hourly output of the new assembly line is higher than that of the old assembly line by at least 300. Can we reject the null at significance level $\alpha = 5\%$?

Note: We will need some of the following z critical values in this problem: $z_{0.1} = 1.2816$, $z_{0.05} = 1.6449$, $z_{0.025} = 1.96$.

Example 7.C: Normal Population, Unknown Variances

Consider a city where pizza restaurants' 10-inch pizzas are with diameters normally distributed with mean μ_X and variance σ_X^2 (unit: inch) and the 12-inch pizzas of theirs are with diameters normally distributed with mean μ_Y and variance σ_Y^2 (unit: inch). The population means and variances above are all unknown. A random sample of 10-inch pizzas with sample size 9 is obtained from city, which is with sample mean $\bar{X} = 9.7$ and sample variance $S_X^2 = 0.0085$, and another random sample of 12-inch pizzas with sample size 7 is obtained, which is with sample mean $\bar{Y} = 11.8$ and sample variance $S_Y^2 = 0.012$.

1. Suppose for THIS PART ONLY that a researcher believes that the population variance of 10-inch pizzas equals that of 12-inch pizzas in the city and wants to test that the diameter of 10-inch pizzas is no more than 2 inches shorter than that of 12-inch pizzas in the city. Write down the null and the alternative hypotheses for him, and check whether we can reject the null at significance level $\alpha = 10\%$.

Note: We will need some of the following t critical values in this problem:

$$t_{0.1,14} = 1.3450, \quad t_{0.05,14} = 1.7613, \quad t_{0.025,14} = 2.1448.$$

2. Suppose that the researcher now wants to test that the population variances of the two sizes of pizzas are equal. Write down the null and the alternative hypotheses for him, and check whether we can reject the null at significance level $\alpha = 5\%$.

Note: We will need some of the following F critical values in this problem:

$$F_{0.05,8,6} = 4.15, \quad F_{0.1,8,6} = 8.10, \quad F_{0.05,6,8} = 3.58, \quad F_{0.1,6,8} = 6.37.$$